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**SUBJECT : PROGRAMMING
FUNDAMENTALS**

ASSIGNMENT 1

QUESTION 1

PROGRAMMING PARADIGMS

PROGRAMMING PARADIGM SIMPLY MEANS A STYLE OR A WAY OF WRITING COMPUTER PROGRAMS OR WAYS OF CODING. IN OTHER WORDS THE PRACTICE OF WRITING A COMPUTER PROGRAM IN DIFFERENT LANGUAGES SUCH AS JAVA, PYTHON IS CALLED PROGRAMMING PARADIGM.

TYPES

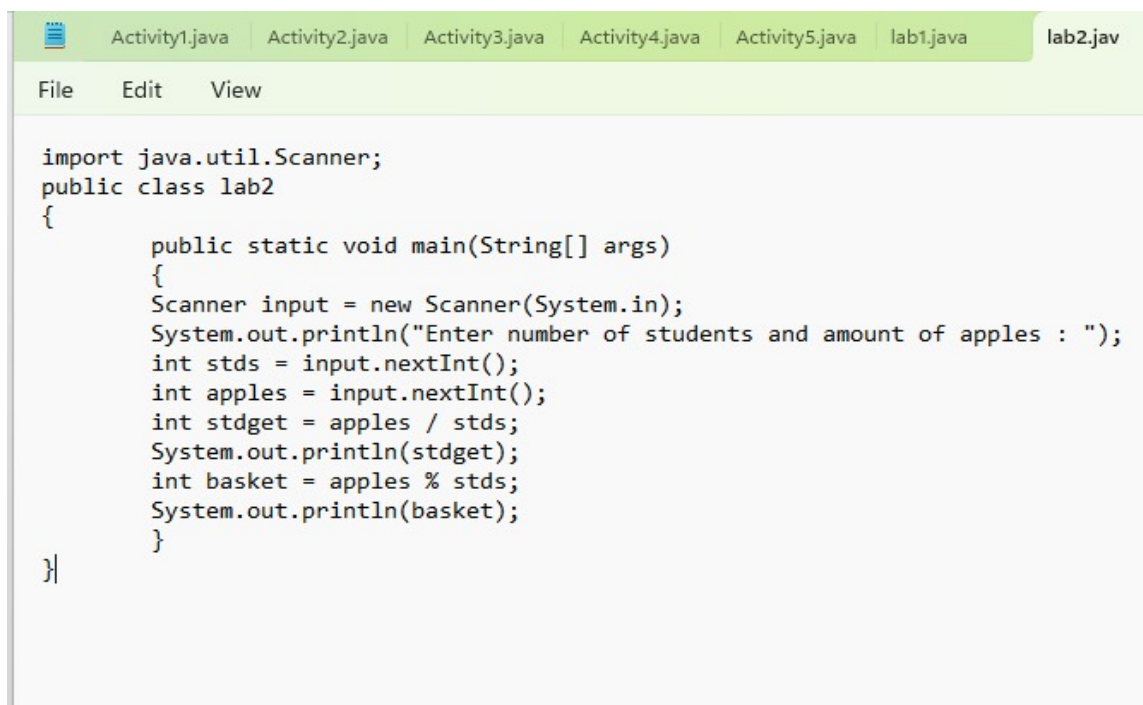
IT HAS TWO TYPES:

1. IMPERATIVE PROGRAMMING
2. DECLARATIVE PROGRAMMING

IMPERATIVE PROGRAMMING

IT IS A PROGRAMMING PARADIGM THAT USES STATEMENTS TO CHANGE A PROGRAM'S STATE, PROVIDING STEP BY STEP INSTRUCTIONS ON HOW TO PERFORM A TASK.

EXAMPLE



```
import java.util.Scanner;
public class lab2
{
    public static void main(String[] args)
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter number of students and amount of apples : ");
        int stds = input.nextInt();
        int apples = input.nextInt();
        int stdget = apples / stds;
        System.out.println(stdget);
        int basket = apples % stds;
        System.out.println(basket);
    }
}
```

DECLARATIVE PROGRAMMING

IT IS A PROGRAMMING PARADIGM WHICH IS FOCUSED ON PRODUCING THE DESIRED RESULTS RATHER THAN SPECIFYING THE STEP BY STEP INSTRUCTIONS WHICH IS CONTROL FLOW TO ACHIEVE IT.

EXAMPLE

A VERY GOOD EXAMPLE OF DECLARATIVE PROGRAMMING IS SQL.

QUESTION 2

(A)

EXAMPLE 1

 *syntax - Notepad
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public class syntax
{
 public static void main(String args[])
 {
 int a = 8;
 int b = 3 //";" missing
 int c = 1: // ":" used instead of ";"
 int avg = (a + b + c) / 3;
 System.out.println(avg);
 }
}

EXPLANATION

1. IN 6TH LINE “;” WAS NOT USED.
2. IN 7TH LINE “:” WAS USED INSTEAD OF “;”

EXAMPLE 2

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labtask5 - Notepad

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```
import java.util.Scanner;
public class labtask5
{
    public static void main(string[] args)//small s used instead of "S"
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter amount of milk produced in the morning : ");
        double amount = input.nextDouble();
        double cost = 0.38;
        double profit = 0.27//missing ";"
        double cartons = amount / 3.78;
        System.out.println("Number of cartons : " + (int)(cartons));
        System.out.println("Cost : $" + amount * cost);
        System.out.println("Profit : $" + profit * cartons);

    }
//missing class closing bracket
```

EXPLANATION

1. IN 12TH LINE “;” WAS NOT USED.
2. IN 20TH LINE CLASS BRACKET WAS MISSING

EXAMPLE 3

 *pay2 - Notepad
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```
import java.util.Scanner;

public class pay2
{
    public static void main(String [] args)
    {
        Scanner keyboard = new Scanner(System.in);
        double hours;
        double rate;
        double pay;

        System.out.print("How many hours did you work?");//missing ";"
        hours = keyboard.nextDouble();
        System.out.print(How much did you get paid per hour?");//missing " at start
        rate = keyboard.nextDouble();

        if(hours <= 40)
            pay = hours * rate;
        else
            pay = (hours - 40) * (1.5 * rate) + 40 * rate;

        System.out.println("you earned $" + pay);
    }
}
```

EXPLANATION

1. IN 13TH LINE THERE IS MISSING SEMICOLON “;”
2. IN 15TH LINE THERE IS MISSING “. AT START DUE TO WHICH MULTIPLE ERRORS ARE SHOWN.

(B)

EXAMPLE 1

```
*SalesTax - Notepad
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import java.util.Scanner;

public class SalesTax
{
    public static void main(String[] args)
    {
        final double TAX_RATE = 0.055;
        double price;
        double tax;
        double total;
        String item;

        Scanner keyboard = new Scanner(System.in);

        System.out.print("Item Description: ");
        item = keyboard.nextLine();
        System.out.print("Item price: $");
        price = keyboard.nextDouble();

        tax = price + TAX_RATE;
        total = price * tax; // INCORRECT FORMULA

        System.out.print(item + "      $");
        System.out.println(price);
        System.out.print("Tax      $");
        System.out.println(tax);
        System.out.print("Total      $");
        System.out.println(total);
    }
}
```

EXAMPLE 2

```
*TemperatureConverter - Notepad
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//3. Fahrenheit to Celsius Converter (Syntax Errors Corrected)
import java.util.Scanner;

public class TemperatureConverter {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter temperature in Fahrenheit: ");
        double fahrenheit = scanner.nextDouble();

        double celsius = (fahrenheit - 32) * 5 / 9; // Incorrect division (should be 5.0 / 9.0)

        System.out.println("Temperature in Celsius: " + fahrenheit); // Incorrect variable name (should be celsius)
    }
}
```

EXAMPLE 3

```
*MultiplicationTable - Notepad
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//4. Multiplication Table (Syntax Errors Corrected)
import java.util.Scanner;

public class MultiplicationTable {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = scanner.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(number + " x " + i + " = " + (number * i)); // "+" used instead of "*"
        }

        scanner.close();
    }
}
```

EXPLANATION

1. IN FIRST INCORRECT FORMULA WAS USED TOTAL SHOULD BE PRICE + TAX
2. IN SECOND DIVISION SHOULB BE 5.0 / 9.0 AND AT LAST CELSIUS SHOULB BE PRINTED NOT FAHRENHEIT.
3. IN THIRD "+" IS USED INSTEAD OF "*" WHICH WONT PRODUCE REQUIRED OUTPUT.

(C)

EXAMPLE 1

```
*Untitled - Notepad
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public class runtime
{
public static void main(String[] args)
{
int a = 3;
b = a / 0; // division by zero reaches infinty
System.out.println(b);
}
}
```

EXAMPLE 2

```
*Untitled - Notepad
File Edit Format View Help
public class runtime
{
public static void main(String[] args)
{
String h = "hello";
int number = Integer.parseInt(text);//wrong typecasting
}
}
```

EXAMPLE 3

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 *Untitled - Notepad

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```
public class runtime
{
public static void main(String [] args)
{
String name = null;
System.out.println(name.length()); //name has no value
}
}
```

EXPLANATION

1. DIVISION BY ZERO IS NOT POSSIBLE SO IT CANT RUN.
2. IN 2ND WRONG TYPECASTING IS PERFORMED HELLO CANT BE CONVERTED TO INT.
3. IN 3RD SINCE NAME HAS NO VALUE SO IT CANT GIVE YOU LENGTH.

(D)

PROGRAM 1

Errors	Solution
Line7: number = "ten"	Should have an integer value i.e. number=10;
Line10: float pi = 3,1416;	Syntax error should be like double pi=3.1416;
Line13: double result ==0;	Invalid initialization should be like double result = 0.0
Line14: (number=5)	Should be like (number ==5)
Line21:("Result is:" +result;	Should be like:("Result is:" +result);

OUTPUT

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>javac error1p.java
C:\Users\shehyar\Desktop\JAVA PROJECTS>java error1p
Result is: 0.0
C:\Users\shehyar\Desktop\JAVA PROJECTS>
```

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PROGRAM 2

Line3: count=1;	Int count=1;
Line4: sum=count + PRIME;(Prime should be initialized)	Int PRIME=2; Int sum= count + PRIME;
Line5: x := 25.67;	Double x = 25.67;
Line6: newNum = count * ONE + 2;	Int newNum=count * 1 + 2;
Line7:sum + count = sum;	sum = sum + count;
Line8: x = x + sum * COUNT;	x = x + sum * count;
Line 9 : “Prime” is written	“PRIME” should be written

OUTPUT

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>javac test.java  
  
C:\Users\shehyar\Desktop\JAVA PROJECTS>java test  
count = 1, sum = 4. PRIME = 2  
  
C:\Users\shehyar\Desktop\JAVA PROJECTS>_
```

QUESTION 3

Line1:multiline comment	/*this program will calculate product of three numbers */
Reserved words	public,class,main,static,args,,int
Special symbol	{}, (), /, *, ;
Predefined(identifier)	System,out,println
Userdefined(identifier)	result,num1,num2,num3,product
Single line comment	//first number,//second number,//third number,//product of number
Output stream	System.out
Input stream	System.in

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QUESTION 4

(A)

```
*Untitled - Notepad
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public class q4
{
    public static void main(String [] args)
    {
        String name = "Java";
        Boolean isPassed = true;
        int product = int a * int b;
        float temperature = 36.6;
        int remainder = num1 % num2;
        double p, q;
        System.out.println("p is" + p + "q is" + q + "" + (p * q) / 2);
        char symbol = '#';
        int v1, v2, v3;
        double resultScore = int score;
        double radius;
        double area = 3.14 * radius * radius;
    }
}
```

(B)

Statement	a=5	b=6	c=7	d=2
a = (++b) * 2 + (c--);	21	7	6	2
c = (a++) - (--d) + b;	22	7	27	1
b = (d--) + (c++) * ++a;	23	622	28	0
d = (--a) + (b++) - (c--);	22	623	27	616

(C)

Statement:	a=3	b=5	c=14.1	sum=0
sum = a + b + (int) c;	3	5	14	22
c /= a;	3	5	4.7	22
b += (int) c - a;	3	6	4	22
a *= 2 * b + (int) c;	48	6	4	22

QUESTION 5

*QN05 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
```

```
public class QN05
{
    public static void main(String[] args)
    {
        Scanner input = new Scanner(System.in);

        int solarrate = 7;
        int companyrate = 60;

        System.out.print("Enter units consumed from solar system: ");
        int solarunits = input.nextInt();

        System.out.print("Enter units consumed from company: ");
        int companyunits = input.nextInt();

        int totalunits = solarunits + companyunits;
        int totalcompany = totalunits * companyrate;
        int actualbill = (solarunits * solarrate) + (companyunits * companyrate);
        int savings = totalcompany - actualbill;

        System.out.println("Total units used: " + totalunits);
        System.out.println("Bill if all from company: Rs. " + totalcompany);
        System.out.println("Actual bill: Rs. " + actualbill);
        System.out.println("Total savings by using solar: Rs. " + savings);

    }
}
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>javac QN05.java
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>java QN05
```

```
Enter units consumed from solar system: 67
```

```
Enter units consumed from company: 89
```

```
Total units used: 156
```

```
Bill if all from company: Rs. 9360
```

```
Actual bill: Rs. 5809
```

```
Total savings by using solar: Rs. 3551
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>
```

QUESTION 6

```
public class QNO6 {  
    public static void main(String[] args) {  
        double weight = 62;  
        double height = 1.68;  
        double result = weight / (height * height);  
        System.out.println("BMI: " + result);  
    }  
}
```

```
C:\Windows\system32\cmd.exe  
Microsoft Windows [Version 10.0.19045.6396]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\shehyar\Desktop\JAVA PROJECTS>javac QNO6.java  
  
C:\Users\shehyar\Desktop\JAVA PROJECTS>java QNO6  
BMI: 21.9671201814059  
  
C:\Users\shehyar\Desktop\JAVA PROJECTS>_
```

QUESTION 7

File Edit Format View Help

```
public class QN072
{
    public static void main(String [] args)
    {
        int salary = 50000;
        int expenses = 37500;
        int laptop = 100000;
        int savings = salary - expenses;
        int months = laptop / savings;
        System.out.println("Ahmed needs" + months + " Months to buy a laptop");
    }
}
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>javac QN072.java
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>java QN072
```

```
Ahmed needs8 Months to buy a laptop
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>
```

QUESTION 8

 QN082 - Notepad

File Edit Format View Help

```
public class QN082
{
    public static void main(String[] args)
    {
        double celsius = 32;
        double fahrenheit = (9 / 5) * celsius + 32;
        double kelvin = celsius + 273.15;
        System.out.println("Fahrenheit : " + fahrenheit);
        System.out.println("Kelvin : " + kelvin);
    }
}
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>javac QN082.java

C:\Users\shehyar\Desktop\JAVA PROJECTS>java QN082
Fahrenheit : 64.0
Kelvin : 305.15

C:\Users\shehyar\Desktop\JAVA PROJECTS>_
```

QUESTION 9

QNO92 - Notepad

File Edit Format View Help

```
public class QNO92
{
    public static void main(String[] args)
    {
        double P = 12000000;
        double annualRate = 0.12;
        double r = annualRate / 12;
        int n = 5 * 12;
        double payment = (P * r) / (1 - Math.pow(1 + r, -n));
        System.out.println("Monthly installment: Rs. " + payment);
    }
}
```

```
Microsoft Windows [Version 10.0.19045.6396]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shehyar\Desktop\JAVA PROJECTS>javac QNO92.java

C:\Users\shehyar\Desktop\JAVA PROJECTS>java QNO92
Monthly installment: Rs. 26693.337221882113

C:\Users\shehyar\Desktop\JAVA PROJECTS>_
```


QUESTION 10

QNO10 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class QNO10
{
    public static void main(String[] args)
    {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter total miles driven per day: ");
        double miles = input.nextDouble();
        System.out.print("Enter cost per gallon of gasoline: ");
        double costgallon = input.nextDouble();
        System.out.print("Enter average miles per gallon: ");
        double mpg = input.nextDouble();
        System.out.print("Enter parking fees per day: ");
        double parking = input.nextDouble();
        System.out.print("Enter tolls per day: ");
        double tolls = input.nextDouble();
        double fuel = (miles / mpg) * costgallon;
        double total = fuel + parking + tolls;
        System.out.println("Total daily driving cost: Rs. " + total);
    }
}
```

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.6396]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shehyar\Desktop\JAVA PROJECTS>javac QNO10.java

C:\Users\shehyar\Desktop\JAVA PROJECTS>java QNO10
Enter total miles driven per day: 34
Enter cost per gallon of gasoline: 45
Enter average miles per gallon: 9
Enter parking fees per day: 345
Enter tolls per day: 32
Total daily driving cost: Rs. 547.0

C:\Users\shehyar\Desktop\JAVA PROJECTS>
```

QUESTION 11

QNO11 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class QNO11
{
    public static void main(String [] args)
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter a number 5 digit");
        int number = input.nextInt();

        int d1 = number % 10;
        number /= 10;

        int d2 = number % 10;
        number /= 10;

        int d3 = number % 10;
        number /= 10;

        int d4 = number % 10;
        number /= 10;

        int d5 = number % 10;

        int reversed = d1 * 10000 + d2 * 1000 + d3 * 100 + d4 * 10 + d5;

        System.out.println(reversed);
    }
}
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>javac QNO11.java
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>java QNO11
```

```
Enter a number 5 digit
```

```
23452
```

```
25432
```

```
C:\Users\shehyar\Desktop\JAVA PROJECTS>
```